**What this workbook covers 本册讲什么**

You cannot study everyone — so the small group you *do* study must carry the weight for everyone else. This booklet builds the vocabulary (target population, representativeness, sampling bias), then walks the four techniques — **stratified, random, opportunity, volunteer** — each with its promise and its failure mode, and drills the two exam shapes: describe-how and define-with-example.

## How to use this booklet 怎么用

1. **Fill it in as you go** — check the answer key (pp. 19–20) after each page.
2. **Page 4–5 (stratified three steps) and page 12 (firstly/secondly/thirdly) matter most.** The 'how' question is a reliable 3-marker — redo it from memory, under time.

## Content map 内容地图

| Page | Topic | What you must be able to do afterwards |
|---|---|---|
| **p.2–3** | Vocabulary 词汇 | Define sampling, target population, representative, sampling bias |
| **p.4–5** | Stratified 分层 | Recite the three steps; compute proportions; state the caveat |
| **p.6** | Random 随机 | Two ways to do it; two ways it breaks |
| **p.7** | Opportunity 机会 | Define with example; ethics pressure point |
| **p.8** | Volunteer 自愿 | Self-selection; why ethical but biased |
| **p.9** | Four compared 对照 | Rank all four on representativeness vs convenience |
| **p.10–11** | Bias diagnosis 诊断 | Spot the bias and state its impact (textbook Activity 4) |
| **p.12–15** | Exam skills 考试 | how-type three steps; define+example; Carlos set; cloze evaluation |
| **p.16–18** | 8-mark & brain dump 演练 | Plan and write the evaluation; dump every term from memory |
| **p.19–20** | Answer key 答案速查 | Mark honestly, then re-write wrong answers in full |

## Revision tracker 复习打卡

☐ p.2–3 vocabulary ☐ p.4–5 stratified ☐ p.6–8 other three ☐ p.9 comparison ☐ p.10–11 bias diagnosis ☐ p.12–15 exam skills ☐ p.16–18 practice

**The whole topic in one sentence 一句话总结**

A sampling technique is the rule that decides **who gets to stand in for everyone**: stratified builds a miniature of the population (most representative), random gives equal chances but leaves the outcome to luck, and opportunity and volunteer buy convenience at the price of **sampling bias** — which limits how far conclusions can be **generalised** back to the target population.

### Fill the gap 填空

**Sampling** is the way in which participants are _____ for a study.

A **target population** is the population of people being _____ by a study.

It is unlikely that a whole population can be studied, so a _____ of the population needs to be gathered using a _____ technique.

### Hospital example 应用

If you are investigating attitudes about health services at a local hospital, the target population will be _____. A sample will be recruited from this target population using a sampling technique.

### The aim 目的

The aim of a sampling method is to select a _____ sample — one that represents the _____ of the population well — so conclusions can be _____ back to explain the behaviour of the target population as a whole.

**中文理解 抽样在解决什么**

研究者想说的是「**全体**怎么想」，手里却只有「**被选中的几十人**」的数据。
中间隔一座桥：**样本像不像总体（representativeness）**。
桥稳 → 几十人的结论推广给全体；桥歪 → 结论只属于那几十人。

## 2 · Sampling bias

### Definition 定义

If the sample gathered is not representative, because of an _____ or _____-representation of a particular type of participant in the sample, a sampling _____ will occur.

### What decides the technique 什么决定技术选择

The technique used will depend on the _____ of research being conducted and the _____ of the participants.

### Quick sort 快速判断

| Statement | Representative / biased / can't tell? |
|---|---|
| A sample of 100 drawn by lottery from the full school roll | |
| A sample of 100 students who all attend the same chess club | |
| A sample where 70% are boys although the school is 50/50 | |

**中文理解 偏差 = 长得不像**

**sampling bias**：样本里某类人**太多或太少**（over-/under-represented）。
检验只问一句：**样本长得像不像总体?** 不像 → 有偏差 → 推广受限。
注意：bias 是**结果**（样本变形），不是方法名——任何技术都可能产出 bias。

## 3 · Stratified sampling: the three steps

### Learn the chain 背链条

**Firstly**, _____ the target population into its _____ (strata).

**Secondly**, calculate the _____ of members of each subgroup.

**Thirdly**, _____ select a _____ number of each subgroup.

### Do the maths 算一算

| Subgroup | % of population | % of a sample of 40 |
|---|---|---|
| Children | 50% | |
| Teenagers | 20% | |
| Adults | 30% | |

### The workplace example 职场例子

Researching stress in a company: find how many staff occupy each role — office clerks, managers, canteen staff, cleaners. There may be _____ clerks than managers, so _____ clerks need to be recruited than managers, to represent the company staff more _____.

**中文理解 三步口诀**

**一分（divide）→ 二算（percentages）→ 三抽（random proportionally）。**

总体里各占几成，样本里就各占几成——**总体的迷你复制品**，所以它是最有代表性的方法。

## 4 · Stratified: the catch

### Weigh it up 权衡

| Strength | Weaknesses |
|---|---|
| | a researcher still needs to ask those selected to take part — and they may _____ to take part |
| most representative sampling method | |
| | time consuming to organise |

### Figure 4.2 in words 图解

Population → _____ (subgroups with percentages) → _____ samples from each stratum. Stratified sampling ensures that each different part of the population is _____ represented in the sample.

### Exam pattern preview 考法预览

"Describe how Crystal might gather a stratified sample of the employees in the office building. (3 marks)" — the full walkthrough is on page 12. Here, just note: the marks live in the **process words** firstly / secondly / thirdly.

**中文理解 金标准也有裂缝**

分层法最像总体，但**被选中 ≠ 会参加**：拒绝的人一多，迷你复制品又变形了。
加上它费时费力——这就是为什么考题爱问「怎么操作」：因为每一步都有分。

## 5 · Random sampling

### Definition 定义

A random sampling technique should ensure that everyone has an _____ chance of being _____.

### Two ways to do it 两种做法

| Method | How it works (fill in) |
|---|---|
| **Computer numbers** | every person is _____ a number; computer-generated numbers select the sample where the numbers _____ |
| **The hat** | place the _____ of every member of the target population into a hat, _____ them, and draw at random |

### Where it breaks 两个裂缝

① You can select an _____ sample at random, too — chance has no memory.
② Even a representative list needs _____ from each participant; those who _____ to take part leave you with an unrepresentative final sample.
③ Like stratified sampling, random sampling is _____ consuming.

**中文理解 随机保证机会，不保证结果**

抽签人人机会均等、研究者手不沾一一听着最公平。
但**运气会歪**（恰好抽出一屋子同款人）+ **拒绝会歪**（不来的都长一个样）。
评价题写这两条裂缝，比背定义更值钱。

## 6 · Opportunity sampling

### Definition 定义

An opportunity sample makes use of participants who are readily _____ by _____ them to take part.

### Two textbook pictures 两个画面

① going to a _____ _____ and asking people to take part; ② investigating _____-by in a street.

### Weigh it up 权衡

| Gain | Costs |
|---|---|
| | limited control over who is _____ |
| quickest and easiest technique | |
| | not everyone has an _____ chance of selection — they may not be _____ at the time |
| | asking people directly can be _____ — it puts _____ on them to participate |

**中文理解 顺手抓人**

谁在 common room、谁路过街头就问谁——**方便是唯一的王牌**。

代价：样本歪（在场的 ≠ 总体缩影）+ 道德压力（当面开口，人难拒绝）。

一句话：**方便换来的，都是代表性欠的债。**

## 7 · Volunteer sampling

### Definition 定义

_____-selected participants can be recruited by placing an _____ in a newspaper or a student common room. Volunteers are self-selecting because they _____ to take part — they are not _____ and asked by a researcher.

### Weigh it up 权衡

| Gain | Costs |
| --- | --- |
| considered a more _____ sampling method — nobody is put on the spot | |
| | the researcher has no _____ over who volunteers |
| | a certain _____ of participant may choose to take part; only those who see the advert and have the _____ respond |
| | can result in sample _____ which limits the _____ of the conclusions |

### Direction matters 方向感

Opportunity: the researcher goes to the people. Volunteer: _____. The flow is reversed — and so is the bias.

**中文理解 贴广告等人报名**
道德上最舒服：**没人被当面施压**。
但报名的人自带滤镜：有空的、看得到广告的、话题相关的——**某一类人**占多数。
对比记忆：**opportunity 人走向研究者？不，研究者走向人；volunteer 才是人走向研究者。**

## 8 · The four compared

### The master table 总表

| Technique | How it works | Main weakness |
|---|---|---|
| **Stratified** | | |
| **Random** | | |
| **Opportunity** | | |
| **Volunteer** | | |

### Rank them 排序

Representativeness (best → worst): _____ > _____ > _____ ≈ _____

Convenience (easiest → hardest): _____ ≈ _____ > _____ > _____

**The pattern 规律**

**Every move away from stratified buys convenience and pays in representativeness.**

考试取用：问 how → 背 stratified 三步；问 define → 方法名 + 例子；问 weakness → 从总表取一条并落到社会心理学场景。

## 9 · Spotting sampling bias I (textbook Activity 4)

### Diagnose each scenario 逐题诊断

| Scenario | Technique | Who is missed? |
|---|---|---|
| (a) Students recruited in the canteen at lunchtime — attitudes to A level reforms | | |
| (b) Volunteers respond to an advert in a women's magazine — career choices | | |

### State the impact 写影响

(a) Impact on conclusions: _____

(b) Impact on conclusions: _____

**中文理解 三连问**

① 谁有机会被选中? ② 谁**没有**机会? ③ 缺的那批人观点会不会**系统性**不同?
会 → 结论推广受限（limited generalisability）——这句就是影响题的万能落点。

## 10 · Spotting sampling bias II

### Diagnose each scenario 逐题诊断

| Scenario | Technique / frame | Who is missed? |
|---|---|---|
| (c) Sample gathered from the telephone directory — health-related behaviour | | |
| (d) Uniform-policy attitudes from a school common room | | |

### State the impact 写影响

(c) Impact on conclusions: _____

(d) Impact on conclusions: _____

### Link to the classics 呼应经典研究

Milgram recruited through a newspaper advert → _____ sampling → self-selected, so his sample over-represented people who respond to adverts. Asch used students available to him → _____ sampling → students only, no older adults. Sampling is one of the fastest evaluation points in any social psychology essay.

## 11 · Exam skills I — the 'how' question

### The rule 规则

If you are asked to describe **'how'** a sampling technique is used, treat this as a _____: _____, _____, _____.
This will ensure you access the available marks.

### The Crystal question 原题

*Crystal has been asked by her employer to send out questionnaires asking opinions about whether the food served in the office canteen is good. There are 100 employees who use the canteen at lunchtime. Describe how Crystal might gather a stratified sample of the employees. (3 marks)*

**Firstly**, Crystal would _____

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**Secondly**, she would _____

---

**Thirdly**, she would _____

---

**中文理解 三步拿三步分**

firstly 分子群（managers / clerks / cleaners）→ secondly 算百分比 → thirdly 按比例随机抽并邀请填问卷。
**过程词就是得分框架——漏掉一步就漏掉一分。**

## 12 · Exam skills II — the 'define' question

### The rule 规则

If asked to _____ a sampling method, you can gain more marks if you give an _____.

### Textbook model 范文填空

"_____ sampling uses participants who are _____ at the time, for example gathering students who are available in the _____ at the time of the study and who agree to take part when _____."

### Write your own 自己写三条

**Random sampling** — definition + example:

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**Stratified sampling** — definition + example:

---

**Volunteer sampling** — definition + example:

---

**中文理解 定义题模板**

**定义 + for example + 具体场景。**没有例子的定义 = 白送的分不要。

## 13 · Mixed practice — name the technique

### Identify and justify 判断并说理由

| Scenario | Technique |
|---|---|
| A researcher numbers every pupil in the school and uses a random number generator to pick 50. | |
| A psychologist stands outside the gym and asks passers-by to complete her survey. | |
| An advert in the common room invites students to sign up for a study on sleep. | |
| A hospital study finds 60% of its patients are over 60, so 60% of the sample is over 60, selected by lottery within that stratum. | |

### Target population spot-check 找总体

| Study | Target population |
|---|---|
| Asking teachers at a school about canteen food | |
| A survey about exercise using a gym membership list | |
| Investigating stress at work in an office building | |

**中文理解 判断抓手**

见「随机数/抽签」→ random；见「站在门口问」→ opportunity；见「广告报名」→ volunteer；见「按比例分子群」→ stratified。

找总体：问的是**谁的意见**，总体就是**那群人**。

## 14 · Exam skills III — the evaluation chain

### The chain 链条

**Technique → bias → limited generalisability.** Fill both cloze models:

"One weakness of opportunity sampling is that the researcher has limited _____ over who is recruited, so not everyone in the target _____ has an equal chance of being selected. The sample may be _____, which limits the _____ of the study's conclusions."

"One weakness of volunteer sampling is that only those who _____ the advert and have the _____ to participate will volunteer, so a certain _____ of participant may dominate the sample. This sample _____ limits the _____ of the conclusions."

### One strength to balance 补一条优点

"One strength of stratified sampling is that subgroups are represented in their true _____, making it the most _____ sampling method, so conclusions _____ well to the target population."

**中文理解 万能链条**

**技术名 → 偏差（谁被多算/漏算）→ 推广受限。**

优点方向倒着走：**按比例 → 最像总体 → 推广最好。**

## 15 · 8-mark planning — sampling in a study

*A psychologist investigates whether people who exercise regularly report lower stress levels, using members of a local gym who are available on Tuesday mornings. Evaluate the sampling decision and its impact on what the study can conclude. (8 marks)*

### Plan it 列提纲

| # | Point | Because + so what |
|---|---|---|
| **P1** | technique = opportunity | |
| **P2** | bias: who is missed? | |
| **P3** | impact on conclusions | |
| **P4** | better alternative + its cost | |

### Write one paragraph in full 写一段

**Scoring self-check 自评**

☐ 报出技术名 ☐ 指出漏掉的人群 ☐ 落到 generalisability ☐ 给了改进方案并说其代价

## 16 · Exam practice — Carlos again (sampling side)

*Carlos wants to test how compliant pedestrians are to 'keep off the grass' signs. He recruits participants using opportunity sampling from pedestrians visiting a local park.*

**(a) Explain how Carlos would recruit his participants. (2)**

**(b) Explain one weakness of this sampling decision for Carlos. (2)**

**(c) Suggest why volunteer sampling would have been more ethical — and one new problem it would create. (2+2)**

**中文理解 提示**

(a) 在公园现场、问当时在场的行人——**谁在场问谁**。
(b) 在场的行人 ≠ 全体行人 → 推广受限（还可以写 direct asking 的道德压力）。
(c) 更道德 = 没人被当面施压；新问题 = 只有看到广告的人报名 → 变一种偏。

## Dump everything 先自测

Cover everything. Write: the four techniques, each with a one-line definition and one weakness; the three stratified steps; the two exam shapes; and every key term you remember:

## The 60-second drill 一分钟自检

- ☐ target population ≠ sample
- ☐ representative = 像总体
- ☐ stratified：分 → 算 → 抽
- ☐ random：等概率，但运气 + 拒绝都会歪
- ☐ opportunity：顺手抓，有施压问题
- ☐ volunteer：自我选择，偏「热心人」
- ☐ bias → limited generalisability
- ☐ how 题 = firstly / secondly / thirdly
- ☐ define 题 = 定义 + for example

### Answer key pp.2–8 答案（上半）

- **p.2** selected; investigated; sample, sampling. People at the hospital. Representative; characteristics; generalised.
- **p.3** over-, under-, bias. Type; availability. Sort: representative (though chance could still skew); biased; biased.
- **p.4** divide, subgroups; percentage; randomly, proportionate. 20 / 8 / 12 out of 40. More; more; fairly.
- **p.5** refuse. Strata; random; proportionally.
- **p.6** equal, selected. Assigned; correspond. Names; shuffle. ① unrepresentative; ② consent, decline; ③ time.
- **p.7** available, asking. Common room; passers-. Recruited; equal, available; unethical, pressure.
- **p.8** Self; advert; choose; approached. Ethical; control; type, time; bias, generalisability. The people come to the researcher.

**Master table answers 总表答案**

Stratified: divide → % → proportional random / refusal + time. Random: equal chance (hat, computer) / chance can misfire, refusals distort. Opportunity: ask the available / unequal chance, pressure. Volunteer: advert, self-select / certain types dominate.

## Answer key pp.9–17 答案（下半）

- **p.9** Ranks — representativeness: stratified > random > opportunity ≈ volunteer; convenience: opportunity ≈ volunteer > random > stratified.
- **p.10 (a)** opportunity; students avoiding the canteen; over-represents canteen users, so conclusions about all students limited. **(b)** volunteer; non-readers of the magazine; one demographic dominates, so career-choice conclusions limited.
- **p.11 (c)** random-ish frame (directory); people without listed numbers; they may differ in age/income/lifestyle → biased health conclusions. **(d)** opportunity; students elsewhere at break; common-room users may share habits → not representative of the school. Classics: volunteer; opportunity.
- **p.12** process; firstly, secondly, thirdly. Firstly divide into subgroups (managers, clerks, cleaners); secondly calculate each subgroup's percentage of the 100; thirdly randomly select the proportionate number and ask them to complete the questionnaire.
- **p.13** define; example. Opportunity; available; common room; asked.
- **p.14** random; opportunity; volunteer; stratified. Teachers at the school; gym members; employees in the office building.
- **p.15** control; population; unrepresentative; generalisability. See; time; type; bias; generalisability. Percentages; representative; generalise.
- **p.17 (a)** go to the park and ask pedestrians available at the time. **(b)** only those present then → unequal chance → unrepresentative → limited generalisability. **(c)** nobody is approached/pressured; but only advert-readers volunteer → a certain type dominates.

**Last look 最后过一遍**

分 → 算 → 抽｜等概率 ≠ 代表性｜顺手抓人有压力｜贴广告来的都是热心人｜**bias → limited generalisability**｜how 题 firstly/secondly/thirdly｜define 题带 for example。